<u>Proposed answers</u>
<u>Question 1</u>

(a) What is the purpose of electroporation? [1]

- Create **transient pores** in the **cell membrane** so as to allow the embryos to pick up the plasmids;;

  ® Answers that seem to indicate that the pores are already present and that electroporation serves to make these pores larger.

(b) Using the information from Figure above as well as your own knowledge,

(i) explain why a strain of bacteria without the lac Z gene was used in this case. [1]

- So that we can be certain that if the bacteria show any β-galactosidase activity, it has to be from the Lac Z gene in the plasmid (that it has taken up) and not due to any inherent Lac Z gene that it possess;;

  OR
- If Lac Z is present in the bacteria, it can code for β-galactosidase which will be able to turn the X-gal blue;
- Therefore cannot differentiate bacteria that has picked up the plasmid with Lac Z gene and those that picked up the plasmid without the Lac Z gene;

NB:
1) Students should avoid using recombinant plasmid and reannealed plasmid without explaining the difference between them.
2) Bacteria that have picked up a foreign piece (in this case the plasmid) of DNA are transformed bacteria regardless of whether the plasmid contains the Lac Z or not.

(ii) describe and explain a procedure that you will carry out that can allow one to distinguish between the two types of bacteria. [3]

- Grow the bacteria in X-gal(substrate);
- Grow overnight in incubator at 37^oC;
- Induce lac operon using IPTG/lactose;
- Lac Z gene codes for β-galactosidase;
- β-galactosidase will hydrolyse X-gal into a blue product;
- Plasmid with intact hobo and lac Z – produce **blue** colonies;
- Plasmid without hobo (and hence no Lac Z) – produce **white** colonies;

(c) Explain the importance of PCR in the above experiment. [2]

- Allows **amplification/** produce many copies of;
- From a minute amount of DNA;
- A **specific sequence**;
- (specific sequence is) hobo element/ hobo-lacZ;
- For further analysis/ allows more accurate determination;

(d) Outline how the PCR products can be analysed to allow the identification of the transposon. [4]

- Run the PCR products on gel electrophoresis;
- Make DNA single stranded/ denature DNA from dsDNA to ssDNA;
- by adding salt and alkali;
- Transfer to nitrocellulose filter using Southern Blot;
- Bake filter at $80^{o}C$ so that DNA is permanently bound;
- Add a single stranded **radioactive** probe;
- which is **complementary to hobo element**;
- **Banding pattern** develops on X-ray film/ autoradiography;

NB:

1) PCR will produce **many copies** of the **same gene** and hence there will not be any difference in the banding pattern due to different fragment size.
2) Students should not be using restriction enzymes to cut PCR products as this is likely to end up cutting the gene that is amplified (i.e if restriction sites are found within the gene).
3) If the DNA is not denatured into single stranded DNA, nucleic acid hybridization with the radioactive probe will not cause the formation of any bands.

(e) Considering all the information that is available,

(i) suggest a possible replacement for Lac Z. [1]
Accept any logical answer eg. gene that interferes with embryonic development/ control the reproductive behavior of the adult insect;;

(ii) discuss the feasibility of controlling the corn earworm through this method. [2]

- Feasibility is low;
- as frequency of *hobo* excision varied only slightly from 1.6%-2.0% *of plasmids* analyzed;
- and only 1.7% of the **adult** harboured the hobo-lac Z insert;
- i.e low percentage of embryos and adults will have the gene being inserted /affected;

(iii) state one advantage of this form of control. [1]

- **specific to the pest** and therefore does not upset ecological balance;;
- does not require the use of pesticide and hence decrease the accumulation of pesticide in the food chain;; Or AVP;;

NB:

1) Students should not just state that the use of pesticide will cause environmental pollution. They should be clearer on what is the effect of pesticide on the environment eg. killing non-targeted animals etc.
2) Some students thought that the gene is inserted into plants instead of the worm and therefore referred to the decrease usage of herbicide.

<u>Question 2</u>

(a) Explain what is meant by ‘genetically modifying rice’. [1]

- Rice that has been genetically altered by the insertion of a modified gene or a gene from another organism;
- Using the techniques of genetic engineering;

(b) Explain the role of the *Hyg* gene in this procedure. [2]

- Helps in selecting for transformed plant cells carrying the recombinant plasmid;
- Transformed cells becomes antibiotic resistant;
- Untransformed cells are susceptible to the antibiotic;
- Transformed cells survive the presence of antibiotic in the selection culture;

(c) A complete mature plant can be developed from callus cells. Suggest a reason to account for this phenomenon. [2]

- Exhibit totipotency;
- A single cell has the ability to divide to produce more cells;
- Retains its ability to use all its genes;
- To produce any type of tissues and eventually the whole plant.

(d) State one problem of transgenes ‘escaping’ into

(i) Natural ecosystem [1]

- Hybridise with wild relatives, thereby making them into superweeds/AVP;; (NB: not the GM crops becoming superweeds.)

(ii) Agricultural ecosystem [1]

- A drop in the quality of conventional and organic crop seed leading to a change in their performance and marketability/AVP;; Or

- Contamination of a conventional maize crop with GM maize may affect the market acceptability of the harvested crop due to reduced quality;;

(e) With reference to Table 2.1 above,

(i) Suggest a reason why the size of the GM plot was increased with an increased in the difference of planting time between GM and non-GM plants. [1]

- Increase the chance of GM pollen still remaining viable so as to effect pollination;;

(ii) State the effect of distance (from the GM plants) on gene transfer. [2]

- Increase distance decrease gene transfer;
- Quote any relevant data;;

(iii) Describe the effect of differences in planting time on gene transfer. [2]

- Increase in differences between planting time, decreases the ability of the GM plants to transfer the gene to non-GM plants;
- Quote Data: Relevant data must be picked to get this mark (eg. 1.02% (=12 m) compared to 0.63% (= 10m) compared to 0.26% (=13m);;

(iii) Suggest a reason for this difference. [1]

- Difference in planting time may result in the pollen no longer being viable when it reaches the non-GM plants;; Or
- Flowers of non-GM plants may not be receptive;;/ AVP;;

(f) Despite concerns over GM crops, researchers believe that these crops can be used to solve the increasing demand for food. Use an example to illustrate how a genetically modified crop can be used in this way. [2]

- named example;
- named gene;
- how to increase yield/ decrease loss;;

Question 3

(a) (i) Discuss the role(s) of the “hematopoietic stem cells”. [1 max]

- Divide mitotically;
- to give rise to a stem cell;
- the other differentiate into various kinds of blood cells;
- multipotent in nature;

(ii) Hematopoietic stem cell is a kind of adult stem cell. Distinguish adult stem cells with the zygotic stem cells. [2]

	Adult stem cells	Zygotic stem cells
Source	Occurs in a differentiated / specialized tissue;	Zygote;
Potency	Multipotent;	Totipotent;

(iii) Explain how it is possible for hematopoietic stem cells to be converted to cell types of a completely different tissue. [1]

- Under appropriate conditions;
- plasticity nature of stem cells may surface / transdifferentiate;

(b) (i) With reference to Figure 3.1 and the information given in the question, explain the different effects of the two CFTR mutations on ion conductance. [4]

- CFTR in plasma membranes needed for conductance of both HCO_3^- and Cl^- ions;
- 33 a.u. for Cl^- ions and 10 a.u. for HCO_3^- ions;
- R117H mutation results in a dysfunctional CFTR in plasma membranes / reduced expression of functional CFTR;
- reduced conductance of both HCO_3^- and Cl^- ions;
- 5 a.u. for both Cl^- and HCO_3^- ions;
- ΔF508 mutation results in absence of the CFTR in plasma membranes;
- negligible or no conductance of both HCO_3^- and Cl^- ions;
- 0.5 a.u. for Cl^- ions and 0 a.u. for HCO_3^- ions;

(ii) Calculate the percentage reduction in chloride ion (Cl^-) transport by cells from heterozygous volunteers, in comparison with cells from volunteers homozygous for the normal CFTR allele. Show your working. [1]

- 33 – 5 = 28 a.u.;
- 28/33 *100 = 84.8%;

(c) (i) For the treatment of hypertension, the antisense approach was adopted. Explain the basis behind the antisense approach. [2]

- express an antisense RNA;
- inhibit mRNA of sense DNA / hybridize with mRNA of sense strand;
- No translation;
- reduce an overexpressed protein / no product formation;

(i) Both the *in vivo* and *ex vivo* approaches had been used for the transference of the antisense genes into somatic cells. Contrast between the *in-vivo* and the *ex-vivo* approach of gene therapy. [2]

Features	In-vivo	Ex-vivo
Cells removal	No cell removed;	Cells removed;
Administration of vectors	Introduce modified vectors directly into specific tissue in body;	Introduce modified vectors to cells in a laboratory setting;

(ii) In the treatment of hypertension using the *in vivo* approach, liposomes might be preferred, over direct injection and viral vectors, for the transfer of the antisense genes. Suggest a reason for this preference for liposomes. [2]

- Multigene disease;
- the cause of the disease lies not in one specific tissue / affecting multiple tissues;
- liposomes fuse with the cell membranes;
- for DNA delivery into cells of various tissues;

Essay Question

<u>Question 4</u>

a) What are the natural and applied uses of restriction enzymes. [4]

Natural:
1. in bacteria as defence;
2. Against foreign DNA/ viral DNA;
3. Cut phosphodiester bonds;
4. Recognizes specific base sequences/ restriction site;
5. Egs of restriction enzymes and the sequence that they cut eg. EcoRI;

(2 max)

Applied uses:
6. allow **formation of recombinant DNA**;
7. to remove/isolate DNA or gene of interest;
8. Prepare plasmid/vector;
9. With the same restriction enzyme;
10. To allow for complementary base pairing through complementary sticky ends;

11. Generate **genomic DNA library**;
12. **DNA analysis**/RFLP analysis;
13. Based on differences in restriction sites/ gain or loss of restriction sites;
14. AVP;

(2 max)

b) Discuss the uses of RFLP analysis. [9]

1. RFLP – refers to the genetic variation that is observed in the size of the DNA fragments that is obtained after the DNA (of different individuals) is digested by a specific restriction enzyme;;

2. **Disease detection**;
3. Direct screening for disease-causing alleles;
4. Due to *loss or gain* of restriction sites/ changes in DNA sequences that affects a restriction site;
5. Generating different size restriction fragments/ sickle cell anaemia – the mutated allele is longer than the normal allele;
6. Use of *RFLP* as proxy for genetic diseases because *linked to disease-causing allele;*
7. *Inherited together* as crossing over is unlikely/ inheritance of the a particular RFLP can indicate the inheritance of the disease causing allele;

8. Allows detection of diseases with late onset/ carriers/ determine genotypes of individuals /prenatal testing (any 2 – max 1m);;

(3 max)

9. **Genomic mapping**;
10. Use of RFLP as genetic markers;
11. The frequency with which two RFLP markers or an RFLP marker and a certain allele for a gene is inherited together is a measure of the closeness of the 2 gene loci on a chromosome /**relative distance** between the 2 gene loci;;
12. Based on recombination frequencies/ frequency of crossing over between 2 RFLP markers/ cross-over values;
13. Can be used to construct linkage map;

(3 max)

14. **DNA fingerprinting/ genetic fingerprinting/ DNA profiling**;
15. Ref to VNTR (variable number of tandem repeats or STR (short tandem repeats);
16. Different **alleles** of VNTR/STR have different lengths;
17. VNTR **highly polymorphic**/ many different alleles;
18. Each individual's **combination is unique**, having inherited one allele for each locus from each parent;
    NB: it is the *combination of alleles* that is unique to an individual, *not the length of the allele* that is unique.
19. Use to identify criminals/ determine paternity/ identify endangered animals etc;; (max 1m)
    ® answers that are not clear eg. genetic fingerprinting use for criminal cases

(3 max)

Award once only – how the different fragments are detected

20. Use of restriction enzymes to cut the DNA at specific sites/ restriction sites;
21. Different sized fragments can be separated by gel electrophoresis;
22. Followed by Southern blotting and nucleic acid hybridization to make the bands visible;

(1 max)

(d) Discuss the importance of tissue culture in the production of genetically modified plants (7).

1. If a new gene is introduced into a plant cell through genetic engineering, the modified cell can be grown into a whole plant;;
2. All cells are genetically identical and carry the inserted gene of interest;;

3. Enable rapid production of large numbers of transgenic plants from one or a few stock plants through asexual means;;
4. As meristematic cells are selected, plant diseases are avoided and thus produce virus free plants;;
5. Prevents the loss of transgenic plants due to sexual reproduction through gamete formation;;
6. Transgenic plants can be produced at any time of the year, put in cold storage and takes up relatively little space;;
7. Reliability and quality control can be achieved since growing conditions can be standardized;;
8. It gives great flexibility in supplying consumer demand because plants can be produced out of season and consumers can buy plants at lower prices than usual;;
9. Transgenic plantlets are light and small in size, thus can be air-freighted and transported easily and cheaply, thus increasing international trade;;
10. Land area needed to grow transgenic plants through plant tissue culture is much less and thus save space;;
11. Prevents transgenic plants from being completely wiped out by changing environmental factors as the plants are always available as stored in cold storage;;

(Any 7 points listed above)